

Going Virtual With OpenVZ

This article is an introduction to OpenVZ, an open source container-based virtualisation solution for Linux.

In computing, virtualisation is the creation of a virtual version of a hardware platform, operating system, storage or network resource. Let's briefly look at each type.

Hardware virtualisation: This provides an environment to create virtual machines and there are three different types—full, partial, and para-virtualisation. In *full virtualisation*, each OS instance and its applications run in a separate VM on top of virtual hardware. The computer system is available as a software construct with the same behaviour as a physical computer system. There are two types of full virtualisation: *bare metal*, by which the hypervisor runs on the underlying hardware, without a host OS; and *hosted virtualisation*, by which the hypervisor runs on top of the host OS (such as Windows or Linux). In *partial virtualisation*, the virtual machine simulates multiple instances of much of an underlying hardware environment—specifically, address spaces—but not the entire OS. In computing, *para-virtualisation* is a technique that represents a software interface to virtual machines that is similar, but not identical to, that of the underlying hardware resources.

Network virtualisation: Here, the physical network is segmented into logical parts to provide network virtualisation by combining network resources such as switches, NICs, firewalls, load balancers, VLANs, storage devices, Ethernet and Fiber Channel, as well as network functionality into a single, software-based network administrative entity. In internal network virtualisation, a host is configured with guest containers to create a virtual network in a box. In external network virtualisation, networks are shared or sub-divided into virtual networks using VLANs and switches.

Storage virtualisation: This is the union of multiple network storage devices into something that appears to be a single storage unit. It provides abstraction on top of various storage techniques, and hence enables flexibility.

Memory virtualisation: This decouples RAM from individual systems in the data centre, and aggregates them into a virtualised memory pool, which can be made available to any computer in the virtual environment.

Software virtualisation: Broadly, we can categorise this into three types:

- *OS-level virtualisation*, by which multiple virtualised environments (multiple isolated user-space instances) can be hosted within a single instance.
- *Application virtualisation* is related to hosting individual applications separated from the underlying OS.
- *Service virtualisation* is emulating the behaviour of dependent components that are needed to execute an application for development or testing purposes.

Data virtualisation: This is the process of abstracting databases, files, etc, with the use of a single data access layer.

Desktop or client virtualisation: This is the virtualisation technology of separating the desktop from the physical machine.

OpenVZ

OpenVZ is a container-based virtualisation solution for Linux. It creates multiple isolated and secure Linux servers known as Virtual Private Servers (VPS) on a single physical machine. Figure 2 depicts its architecture. Each container or VPS performs and executes instructions exactly like a stand-alone server. A VPS has root access, users, processors, memory, IP addresses,